

Lecture 1: (Artificial) Intelligence, Learning, Language, and Meaning

Language models now write programs, prove small lemmas, summarize papers, and hold conversations that many people find hard to tell apart from human ones. How these systems work is mostly well understood at the level of their parts: a probability distribution over strings, parameterized by a large network, fitted to text, adjusted with demonstrations and rewards, and sampled from at inference time. Understanding why they work, what they can and cannot do, and how to reason about them is much less settled. This course studies the whole stack from first principles, from a theoretical computer science viewpoint, and builds a small working language model along the way.

This first lecture has four parts. It opens with a short history, from the first mathematical models of computation to today's language models, and states what the course will build. It then fixes the vocabulary of artificial intelligence, machine learning, and deep learning. The middle of the lecture takes up four old questions that these systems make newly concrete: what intelligence is, what it means to learn, what language is, and what meaning is. Those sections are more discursive than the rest of the course; they are meant to frame what follows, not to settle anything. The lecture ends with its hands-on segment, a walk through the automatic-differentiation engine that trains every neural model in the course, and with a map of how the pieces fit together. Section 9 collects the formal results behind the discussion for readers who want them; the lecture itself does not cover them.

1. From computation to language models

Tools that extend what people can do

People have long built tools that extend what they can do. Some extend physical ability: the control of fire, agriculture, the wheel, industrial power. Others extend what can be remembered and shared. Writing and numerals store information outside a single mind, printing copies it, and electrical communication moves it across distances almost instantly. Computers belong to the second family and add a new capability: they transform information by following a procedure. Language models are a recent and unusual step in that line, because they make a wide range of tasks accessible through ordinary language.

Computation and artificial intelligence

Year	Landmark	What it introduced
1936	Turing's model of computation	Mechanical symbol manipulation becomes a precise mathematical object
1946-1947	ENIAC and the transistor	Programmable electronic computation, and the device that later made it small and cheap
1950	Turing's imitation game	An operational question about observable behavior in place of "Can machines think?"
1955-1956	The Dartmouth proposal and workshop	Artificial intelligence named as a research program
1958	Rosenblatt's perceptron	A decision rule learned from examples by adjusting numerical weights
1959	Samuel's checkers program	A checkers program that improves by playing; an early use of the term "machine learning"
1986	Backpropagation popularized	Multilayer networks that learn intermediate representations
1997	Deep Blue defeats Kasparov	Search, a hand-built evaluation function, and special-purpose hardware win a match against the world chess champion
2012	AlexNet on ImageNet	Learned features, a large labeled dataset, and GPUs outperform engineered features in image recognition
2014	Generative adversarial networks	Neural networks that generate realistic images
2016	AlphaGo defeats Lee Sedol, 4-1	Learned policy and value networks combined with search

These are landmarks, not eras. Artificial intelligence, machine learning, deep learning, and generative modeling are overlapping fields that developed side by side, and search and learning have been combined throughout. Deep Blue and AlphaGo both search; they differ mainly in where their knowledge of the game comes from, a contrast Section 4 returns to.

Language models

Year	Landmark	What it introduced
2017	The Transformer	Attention-based sequence modeling: the architecture this course implements
2020	GPT-3	Tasks specified by a prompt and a few examples, with the weights held fixed
2022	ChatGPT	Post-training and a conversational interface make a model usable as an assistant
2023-2024	GPT-4 and GPT-4o	Images as input, and then one model announced for text, audio, and images
2024-2025	OpenAI o1 and DeepSeek-R1	Post-training for reasoning, and more computation spent at inference time
2025	Gemini Deep Think at the International Mathematical Olympiad	Gold-medal standard: five of six problems solved, 35 of 42 points, graded by the competition's coordinators

The selection stops at 2025 and is not a survey of releases. It is meant to show what changed: language became a general interface for specifying tasks, models learned to follow instructions, their inputs widened beyond text, and their performance on problems with checkable answers rose sharply. Claims of this kind should be read together with their evidence. The Olympiad result concerns an advanced research system on a specified competition, graded by independent judges; it does not show that a chatbot's mathematics is always correct. Asking what an output demonstrates, and how it was checked, is part of the working understanding this course aims for.

2. What we will study and build

The central object

A generative model produces structured objects: a paragraph, a program, an image, a sequence of actions. This course concentrates on language, for two reasons. Language models are the systems that changed the field between 2018 and today, and language is the setting in which the mathematics is richest: it has discrete symbols, recursive structure, compositional meaning, dependence on context, and a long history in logic, linguistics, and the theory of computation.

The object at the center of the course is a probability distribution over strings. Fix a finite vocabulary $\mathcal{V}$ of tokens (characters, words, or pieces of words) together with a special end-of-sequence token EOS. A language model assigns a probability to every finite string $x_1x_2\cdots x_T$ by predicting one token at a time:

$$q_\theta(x_{1:T}, \text{EOS}) = \prod_{t=1}^{T+1} q_\theta(x_t \mid x_{<t}), \quad x_{T+1} = \text{EOS}.$$

Here $x_{<t}$ denotes the tokens before position t , and θ collects the model's parameters. The identity on the right is just the chain rule of probability; the modeling content lies entirely in how each conditional distribution $q_\theta(\cdot \mid x_{<t})$ is computed and how θ is chosen. Almost everything in the course is a statement about this object. Weeks 2 and 3 are about how the conditionals are computed: first with counts and grammars, then with attention. Week 4 is about how θ is fitted at scale. Weeks 5 through 7 are about how the fitted distribution is changed so that sampling from it produces a helpful assistant rather than a plausible continuation of web text. Week 8 is about how the distribution is used: decoded, searched, served, and evaluated.

What we will build

The course has two aims, and they support each other. The first is a **mathematical understanding** of this object: its representations, probabilities, and objectives; the algorithms that fit it and use it; their computational costs; and the assumptions behind whatever guarantees are available. The second is a **working understanding**: the ability to implement each component, prepare data, train a model, inspect its predictions and errors, fine-tune it, probe its behavior, and evaluate what it produces.

The second aim is served by one small codebase, `tiny1m`, which gains a component each week. It starts in this lecture with an automatic-differentiation engine (Section 8). It then grows through count-based language models, a Transformer, pretraining, supervised fine-tuning that turns the pretrained model into a small chatbot, post-training with preferences and rewards, and efficient generation and evaluation. The count models are baselines within the same project, not an early version of the Transformer; what the Transformer replaces is the computation of the conditional probabilities. Every model is small enough to train on a laptop in minutes to an hour. At that scale the quality of its output is modest, but the mechanisms of the large systems are all present and can be inspected, changed, and tested.

“From scratch” has a specific meaning. We implement the models and their training methods from tensor operations and automatic differentiation, and we write the differentiation engine itself once, in this lecture’s hands-on segment, so that it is not a black box when the course starts relying on PyTorch’s. We do not reimplement numerical libraries, and from Week 2 on we use PyTorch tensors without higher-level model libraries.

The eight weeks

Week	Lecture	Central development	We build
1	(Artificial) Intelligence, Learning, Language, and Meaning	History and vocabulary; intelligence, learning, language, and meaning	A tensor autograd engine; logistic regression and a small network
2	Language Modeling: From Grammars and Counts to Neural Networks	Formal languages and stochastic sources; n-gram models, smoothing, and perplexity; hidden Markov models; distributional semantics; the first neural language models	Count-based language models, a sampler, and an evaluation harness; count-based word vectors
3	Attention and Transformers: Computation and Limits	Recurrent networks and their limits; attention; the complete Transformer computation, its costs, and what it can represent	A recurrent baseline and a decoder-only Transformer
4	Pretraining: Data, Optimization, and Scale	Tokenization, data, objectives, optimization, scaling laws, and training systems	A BPE tokenizer, a training loop, and a small scaling experiment
5	From Base Models to Assistants	Completion, prompting, and in-context learning; the gap between predicting text and following instructions	A chatbot, by supervised fine-tuning with LoRA
6	Reinforcement Learning for Sequence Generation	Bandits, MDPs, value functions, policy gradients; text generation as a sequential decision problem	Value iteration and policy gradients, then rewards for the chatbot
7	Post-Training: Demonstrations, Preferences, and Rewards	Supervised fine-tuning, reward models, RLHF, DPO, and verifiable rewards	A reward model, DPO, and GRPO on arithmetic
8	Inference: Decoding, Test-Time Compute, Serving, and Evaluation	Decoding, test-time computation, serving, and evaluation	Sampling methods, a key-value cache, speculative decoding, and evaluation with confidence intervals
9 (optional)	Alignment and Language Model Agents	Specifying and evaluating desired behavior; tools, retrieval, and agent loops	A tool-use loop

The order follows the dependencies between ideas. Traditional language modeling comes first because the questions it raises (sparsity, evaluation, what a distribution over strings is) remain the questions of the neural era. Attention comes before scale because scaling is a statement about a fixed architecture. The general theory of reinforcement learning gets a lecture of its own before its use in post-training, because the post-training objectives are much easier to understand once the underlying decision problem is clear.

The point of view

The course is presented from a theoretical computer science viewpoint. That viewpoint is less a set of results than a habit of asking particular questions:

- **What is computed?** A trained Transformer computes a specific function; decoding computes another; a chat system wraps both in a larger program. Being precise about the function is the first step to reasoning about it.
- **With what resources?** Training compute, memory, the cost of one forward pass, and the cost of generating a complete answer are different quantities, and they scale differently.
- **From what information?** A learner can only extract what its data contain. Much of the course is about which information each stage of training supplies: raw text, demonstrations, comparisons, or checkable outcomes.
- **With what guarantee?** Some claims about language models are theorems, some are measurements, and some are conjectures supported by a handful of experiments. Keeping them apart is half the work.

A recurring instance of this habit is the separation of **expressivity**, **optimization**, and **generalization**. A model class may be able to express a computation without any efficient procedure finding it; a procedure may find parameters that fit the training data without the result working on new inputs. Much confusion about what language models “can do” comes from running these three questions together.

3. Terminology

The opening used words such as intelligence, learning, and model freely. This section fixes their meanings before going further. It is a map, not a course: Foundations 1 of the Background Notes on Machine Learning and AI develops the same vocabulary with more examples, and Section 10 lists where each basic topic is treated in full.

AI, machine learning, and deep learning

Artificial intelligence is the study of computational systems that perform tasks associated with intelligence: perception, reasoning, language, planning, and action. Its methods include search (exploring a space of possibilities, as in game playing and planning), logic (representing knowledge and deriving consequences), probabilistic reasoning (representing uncertainty with probability distributions and updating them with evidence), and learning. A practical system usually combines several of these: a chat assistant is a learned model embedded in a program that manages conversation, may search over candidate answers, and may call tools.

Machine learning is the study of procedures that improve with experience. Tom Mitchell’s definition is still the most precise short one: a program learns from experience E with respect to a class of tasks T and a performance measure P if its performance on tasks in T , as measured by P , improves with E . For a language model, T might be predicting the next token of web text, E a corpus of documents, and P the average log probability assigned to held-out text.

Deep learning is machine learning with models built as compositions of many trainable transformations, usually neural networks, so that the model learns intermediate representations of its input along with its predictions. The depth is the length of the composition.

Statistics shares many of these methods: estimation, uncertainty, experimental design, and prediction. Neither field owns inference or prediction; the difference is mostly one of emphasis, with statistics making the data-generating assumptions explicit and machine learning emphasizing flexible models and computation.

Sources of learning signal

Learning procedures differ in where their training signal comes from.

- **Supervised learning** uses examples of inputs paired with targets, (x_i, y_i) : images with labels, or instructions with demonstrated answers.
- **Unsupervised learning** receives inputs without targets and models their structure: a density, clusters, or a low-dimensional representation.
- **Self-supervised learning** constructs targets from the data itself. A language model predicts each token from the ones before it; a masked model predicts hidden tokens from the visible ones. The provenance of the target differs from supervised learning, but the training calculation is an ordinary supervised loss.
- **Reinforcement learning** learns from rewards obtained by acting. The learner's actions affect what it observes next, and rewards may arrive long after the actions that earned them.

These categories describe the source of the signal, not the architecture. A single Transformer is pretrained with self-supervision, finetuned with supervision, and adjusted with rewards, as the course will describe week by week.

Models, algorithms, and parameters

A **model** is a mathematical object used to represent a relationship or a distribution. A **learning algorithm** maps data to a fitted model; a separate computation then evaluates the fitted model on new inputs. The **representation** determines what information about an input reaches the model: a bag of words discards order, a token sequence keeps it. An **architecture** specifies the computation and how parameters are shared. **Parameters** are the quantities adjusted by learning, such as the weights of a network. **Hyperparameters** govern the model family or the learning procedure, such as the learning rate, the model width, or a regularization strength, and are chosen during development.

For example, represent an email by a vector x of features, with label $y = 1$ for spam and $y = 0$ otherwise. Logistic regression models the conditional probability of spam as

$$q_w(1 | x) = \sigma(w^\top x), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

The vector w holds the parameters, $w^\top x$ is called the **logit**, and a decision rule such as “flag as spam if $q_w(1 | x) > 1/2$ ” turns the probability into an action.

Generative and discriminative models

A **generative** classifier models the joint distribution of inputs and labels, often as $q(x, y) = q(x | y)q(y)$, and classifies with Bayes' rule,

$$q(y | x) = \frac{q(x | y)q(y)}{\sum_{y'} q(x | y')q(y')}.$$

A **discriminative** classifier models $q(y | x)$ directly, as logistic regression does. Ng and Jordan, 2001. The word “generative” in generative AI is broader: it refers to models from which new objects can be sampled. A language model is generative in this sense even though each of its conditionals is a classifier over the next token.

Training, generalization, and evaluation

A **loss** measures the cost of a prediction. For probabilistic predictions the standard choice is the log loss $\ell(\theta; x, y) = -\log q_\theta(y | x)$, which penalizes assigning low probability to what actually happened. If examples are drawn from a population distribution p_* , the quantity we care about is the **population risk** $R(\theta) = \mathbb{E}_{(x,y) \sim p_*} [\ell(\theta; x, y)]$. We can only compute the **empirical risk** on the n training examples,

$$\widehat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; x_i, y_i),$$

and **empirical risk minimization** chooses θ to make it small, often with a regularization term added.

Generalization is the relationship between the two. **Overfitting** occurs when the empirical risk falls without a corresponding fall in the population risk. The Occam bound of Section 5 is the simplest example of a generalization guarantee: with enough examples relative to the size of the hypothesis class, empirical and population performance are close.

In practice, generalization is estimated rather than proved. Training data fit the parameters; **validation data** guide development choices such as the architecture, learning rate, and stopping time; **test data** estimate performance once those choices are fixed. The separation concerns information flow, not filenames: a test set that has been used to choose a model has become part of development. For text, overlapping windows or duplicated passages can leak information across a split even when the split looks clean, so splits are made by document, and duplicates are removed.

Training changes a model’s parameters; running the trained model with its parameters fixed is called **inference**. For language models, training itself has stages: **pretraining** on a large corpus of text, and **post-training**, which adapts the pretrained model with demonstrations, comparisons, or rewards. Section 8 summarizes the stages and the signal each one uses.

The word **inference** is badly overloaded, and it is worth separating its meanings once:

Term	Meaning	Example
Statistical inference	Reasoning from data to unknown quantities, including their uncertainty	A confidence interval for a model’s accuracy
Probabilistic inference	Computing a quantity within a specified model	The posterior probability of spam given the words of an email
Inference (deployment)	Running a trained model	Generating a response to a prompt

4. Intelligence

Turing’s question

In 1950 Alan Turing proposed to replace the question “Can machines think?”, which he considered too ill-defined to discuss, with a game. An interrogator converses by text with two hidden parties, a person and a machine, and must decide which is which. The machine does well if the interrogator is wrong about as often as with two people. Turing predicted that by about the year 2000 a machine with roughly 10^9 bits of storage would play well enough that an average interrogator would have no more than a 70 percent chance of a correct identification after five minutes. Turing, *Computing Machinery and Intelligence*.

Two features of the paper matter for this course. First, the test is behavioral and linguistic: intelligence is to be judged by conversation, which is exactly the capability that language models now exhibit. Second, most of the paper is not about the game at all but about objections to machine intelligence, and its final section, “Learning Machines,” argues that the practical route to a thinking machine is not to program adult competence directly. “Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child’s?” The child machine would then be educated. Turing also answers Lady Lovelace’s objection that a machine “has no pretensions to originate anything” and can only do what it is told, by observing that a learning machine’s behavior is not fully anticipated by its programmer. Seventy years later, the systems that come closest to passing his test were built in roughly the way he suggested: a general learning procedure, a great deal of experience, and comparatively little explicit programming.

The imitation game is a poor definition of intelligence, and Turing did not offer it as one. It is anthropocentric, it rewards deception, and people attribute understanding on thin evidence: Joseph Weizenbaum’s 1966 ELIZA program, a few pattern-matching rules that turned a user’s statements back into questions, left some of its users convinced that it understood them. Weizenbaum, 1966. Its lasting value is methodological: it replaces a metaphysical question with an operational one about observable behavior. Much of this course does the same thing at a smaller scale, replacing “does the model understand?” with “what does the model compute, and how does it perform on a specified distribution of tasks?”

Symbols, networks, and the Dartmouth conjecture

The 1955 proposal for the Dartmouth summer workshop, which named the field, stated its founding conjecture: “every aspect of learning or any other feature of intelligence can in principle be so precisely

described that a machine can be made to simulate it.” Its list of topics included how a computer can be programmed to use a language, neuron nets, self-improvement, abstraction, and randomness and creativity. McCarthy, Minsky, Rochester, and Shannon, Dartmouth proposal. Almost every item on that list is a topic of this course.

Two broad programs developed from it. The symbolic program held that intelligence consists in manipulating symbols according to rules. Allen Newell and Herbert Simon stated its strongest form as the physical symbol system hypothesis: “a physical symbol system has the necessary and sufficient means for general intelligent action.” Newell and Simon, 1976. Theorem provers, planners, expert systems, and hand-written grammars belong to this program. The connectionist program held that intelligence emerges from large numbers of simple units adjusting their connections. It begins with Warren McCulloch and Walter Pitts’s 1943 logical model of neurons, which showed that networks of threshold units can compute any Boolean function, and with Frank Rosenblatt’s perceptron, a single threshold unit that learns its weights from labeled examples. McCulloch and Pitts, 1943; Rosenblatt, 1958. Marvin Minsky and Seymour Papert’s 1969 book *Perceptrons* proved that single-layer perceptrons cannot compute predicates such as parity or connectedness from local features, and the lack of an effective training method for multilayer networks stalled the program until backpropagation was popularized in the 1980s.

A modern language model is a descendant of the connectionist program: a network of simple units whose connections are learned. Yet it operates on discrete symbols, produces symbol sequences, and can be prompted to carry out symbolic procedures such as arithmetic or program execution, sometimes reliably and sometimes not. Whether it manipulates symbols in the sense of Newell and Simon, and whether that question even has a sharp answer, is one of the live debates this course equips you to think about.

Search and learning: Deep Blue and AlphaGo

Game playing has served as a test bed for AI since the 1950s because success in it is unambiguous. Two landmarks show how differently success can be achieved. IBM’s Deep Blue, which defeated the world chess champion Garry Kasparov in a six-game match in 1997, combined a deep search of the game tree with an evaluation function built by hand from thousands of features of a chess position, running on 480 custom chips that together examined on the order of a hundred million positions per second. Campbell, Hoane, and Hsu, 2002. Its knowledge of chess came from people; its strength came from search. Go resisted this approach: a position has roughly 250 legal moves rather than about 35, and nobody could write a good evaluation function by hand. DeepMind’s AlphaGo, which defeated Lee Sedol 4-1 in 2016, learned the functions it needed. A policy network, trained first to predict the moves of expert players and then improved by playing against itself, proposed promising moves; a value network, trained on positions from self-play, estimated who was winning; and a tree search combined the two. Silver et al., 2016.

Both systems search. They differ in where their knowledge comes from, and the same division runs through this course. A pretrained language model is a learned policy over next tokens. Post-training with rewards (Weeks 6 and 7) improves it much as self-play improved AlphaGo’s policy, and the test-time computation of Week 8, which samples many answers, verifies them, or searches over steps of reasoning, is search wrapped around a learned model.

Definitions and measures

There is no agreed definition of intelligence, but there are useful families of definitions. The standard textbook of the field, Stuart Russell and Peter Norvig’s *Artificial Intelligence: A Modern Approach*, sorts them along two axes: whether they concern thinking or acting, and whether they measure against human performance or against an ideal of rationality. The book itself adopts the rational-agent view: an agent perceives its environment and acts so as to maximize the expected value of a performance measure, given what it has perceived. This view is convenient for engineering because it turns intelligence into an optimization problem, and it is the view behind reinforcement learning in Week 6.

Shane Legg and Marcus Hutter pushed the rational-agent view to a formal limit: the universal intelligence of an agent is its expected performance across all computable environments, each weighted by its simplicity, so that simple environments count most. Legg and Hutter, 2007. The definition captures a common intuition, that intelligence is the ability to do well across many environments with a

bias toward simple ones. It is also incomputable, so it cannot serve as a benchmark; Section 9 gives the formula.

François Chollet objects that measuring skill at tasks, even many tasks, conflates intelligence with the knowledge and priors that were used to build the skill. A system can buy performance on any fixed set of tasks with enough training data. He proposes instead to measure intelligence as *skill-acquisition efficiency*: how quickly a system acquires new skills over a scope of tasks, relative to its priors, its experience, and the difficulty of generalization. Chollet, 2019. The distinction matters for language models, which are trained on a large fraction of the written record: their performance on a benchmark says something about their skill, but by itself very little about how efficiently they learn.

Behavior, understanding, and the bitter lesson

John Searle’s Chinese room argument attacks the behavioral view directly. A person who speaks no Chinese sits in a room with a rulebook for manipulating Chinese characters. Questions in Chinese come in; by following the rules, the person sends out answers that native speakers find perfectly sensible. Searle argues that the person does not understand Chinese, that neither does the room, and therefore that running the right program is not sufficient for understanding. Searle, 1980. The argument has many replies, the most common being that the system as a whole, rather than the person, is the relevant candidate for understanding. Whether the argument applies to language models is itself disputed. A language model resembles the room in being a fixed procedure that maps input strings to output strings, with its competence derived from text rather than from contact with the world. It differs in that its procedure was learned rather than written, and in that no single component plays the part of the person. The analogy settles nothing, but it sharpens a question that Section 7 takes up in the form it has for language models: can meaning be learned from form alone?

Two further observations from the history of AI shape the course. The first is Moravec’s paradox, named after Hans Moravec: it proved comparatively easy to make computers perform well at tasks that humans find intellectually demanding, such as chess or symbolic integration, and very hard to give them the perception and mobility of a one-year-old child. The skills that seem hard to us are recent in evolutionary terms and small in their demands; the skills that seem effortless are old and enormous. Language models add a twist: fluent language, which humans find effortless, turned out to be learnable from text at scale, while reliable multistep reasoning, which humans find effortful, remains harder for them.

The second is Richard Sutton’s “bitter lesson”: across seventy years of AI research, general methods that exploit increasing computation, chiefly search and learning, have eventually beaten methods that build in human knowledge about the domain, often by a large margin. Sutton, 2019. The progression from hand-written grammars to n-gram models to neural language models trained on trillions of tokens is one of the clearest instances. The lesson is empirical, not a theorem, and it does not say that structure is useless: the Transformer is itself a carefully designed structure, and much of Weeks 3 and 4 is about which structural choices matter at scale.

5. Learning

The problem of induction

Every learning system generalizes from what it has seen to what it has not. David Hume observed in 1748 that no argument from experience can establish that unobserved cases will resemble observed ones without assuming the very regularity it sets out to prove. Hume, *An Enquiry Concerning Human Understanding*. Any justification of induction must therefore rest on assumptions about the world that the data alone cannot supply.

Machine learning has a precise version of this observation, usually called a no-free-lunch theorem. Consider binary classification on a finite domain of N points, and suppose the true labeling is chosen uniformly at random from all 2^N functions. After observing the labels of any subset of the points, the label of every unobserved point is still a fair coin, independent of everything seen. Every learning algorithm, however clever, then has expected error exactly one half on unseen points. Wolpert, 1996. Learning is possible only because real problems are not like this: the functions we care about are smooth, compositional, or otherwise simple, and a learner succeeds by preferring such functions. That preference is its **inductive bias**. In a language model the inductive bias lives in the tokenizer, the architecture, the

optimizer, the order of the data, and the objective, and much of the course is about what each of these contributes.

From the perceptron to deep learning

The learning systems of this course descend from a simple procedure. Rosenblatt's perceptron classifies an input $x \in \mathbb{R}^d$ by the sign of $w^\top x$, and it learns from its mistakes: whenever it misclassifies an example (x, y) with label $y \in \{-1, +1\}$, it updates $w \leftarrow w + yx$. If some unit vector u separates the data with margin γ , meaning $yu^\top x \geq \gamma$ for every example, and every input has norm at most R , then the perceptron makes at most $(R/\gamma)^2$ mistakes, however many examples it sees (Novikoff, 1962; Chapter 9 of Shalev-Shwartz and Ben-David). The bound depends on the geometry of the data, not on the dimension or the number of examples.

A single threshold unit can only draw a linear boundary, the limitation Minsky and Papert made precise. Stacking units into layers removes the limitation, but the perceptron's rule cannot train a multilayer network, because a mistake at the output does not say which hidden unit is to blame. The answer, popularized by David Rumelhart, Geoffrey Hinton, and Ronald Williams in 1986, is to make every unit differentiable and to follow the gradient of a loss. Rumelhart, Hinton, and Williams, 1986.

Neural networks and gradient descent

A feedforward neural network composes affine maps with fixed nonlinear functions: a layer computes $h \mapsto \phi(Wh + b)$, where W and b are parameters and ϕ is applied coordinatewise, for example $\phi(z) = \max(0, z)$. Without the nonlinearity a stack of layers would collapse to a single affine map. With it, networks with one hidden layer of sufficient width can approximate any continuous function on a compact set to any accuracy. Cybenko, 1989; Hornik, 1991.

The universal approximation theorem is a statement about expressivity only, and a TCS reader should resist over-reading it. It says nothing about how large the network must be, how many examples are needed to find a good approximation, or whether gradient descent will find it. Learning parity functions of a few hidden bits from random examples, for instance, is easy to express and notoriously hard for gradient methods. The success of deep learning is an empirical fact about how expressivity, optimization, and generalization interact on natural data, and the theory explaining it is incomplete. LeCun, Bengio, and Hinton, 2015.

Architectures differ in how they share parameters, which encodes an assumption about the data. A convolutional network reuses the same filters at every position of an image, assuming that local patterns matter wherever they occur. A recurrent network applies the same update at every position of a sequence, carrying a hidden state that summarizes the past. A Transformer lets every position of a sequence read from every earlier position through attention, with the same parameters at every position. Weeks 2 and 3 develop the last two in detail.

Networks, like most models in this course, are fitted by gradient methods. **Gradient descent** repeatedly moves the parameters against the gradient of the loss, $\theta \leftarrow \theta - \eta \nabla \widehat{R}(\theta)$, with a learning rate $\eta > 0$. **Stochastic gradient descent** replaces the full gradient with an unbiased estimate computed from a small random subset of examples, a **minibatch**, trading noise for much cheaper steps. Variants such as momentum, Adam, and AdamW adapt the step using the history of gradients; Week 4 examines them.

For logistic regression with log loss, the gradient on one example is $(\sigma(w^\top x) - y)x$: the prediction error times the input. Starting from $w = (0, 0)$ with $x = (2, 1)$ and $y = 1$, the predicted probability is $1/2$, the loss is $\ln 2 \approx 0.693$, and the gradient is $-\frac{1}{2}(2, 1)$. One step with $\eta = 0.1$ gives $w = (0.1, 0.05)$, and the predicted probability of the correct label rises to $\sigma(0.25) \approx 0.562$. The step improved the prediction on the example it was computed from. Whether it helps on other examples is a separate question, which only held-out data can answer; in Exercise 5 the same step makes a second example worse.

The gradient itself is computed by **backpropagation**, the chain rule applied backwards through the network. Section 8 builds an engine that does this for any computation made of differentiable operations.

Deep learning, the training of many such layers end to end on large datasets, displaced engineered features first in speech and image recognition, most visibly when AlexNet won the 2012 ImageNet competition by a wide margin, and then in language. Krizhevsky, Sutskever, and Hinton, 2012.

Learning as a computational problem

Leslie Valiant turned learning into a problem about algorithms and resources. In the probably approximately correct (PAC) model, there is an unknown distribution $\mathcal{D}$ over an input domain X and an unknown target concept $c : X \rightarrow \{0, 1\}$ drawn from a known class $\mathcal{C}$. The learner receives m independent examples $(x_i, c(x_i))$ with $x_i \sim \mathcal{D}$ and must output a hypothesis h whose error

$$\text{err}_{\mathcal{D}}(h) = \Pr_{x \sim \mathcal{D}}[h(x) \neq c(x)]$$

is at most ε , with probability at least $1 - \delta$ over the sample. The class is efficiently PAC learnable if some algorithm does this for every $\mathcal{D}$ and every $c \in \mathcal{C}$, using a number of examples and an amount of time polynomial in $1/\varepsilon$, $1/\delta$, and the size of the problem. Valiant, *A Theory of the Learnable*, 1984.

The definition makes both halves of Hume’s problem explicit. The distribution $\mathcal{D}$ is the assumption that the future resembles the past: test inputs come from the same distribution as training inputs. The class $\mathcal{C}$ is the inductive bias: the learner knows the target is one of a restricted set of functions. Given these, learning becomes possible, and its cost can be measured.

The simplest positive result shows how the cost can be measured. If the learner searches a finite hypothesis class $\mathcal{H} \supseteq \mathcal{C}$ and outputs any hypothesis that agrees with every training example, then

$$m \geq \frac{1}{\varepsilon} \left(\ln |\mathcal{H}| + \ln \frac{1}{\delta} \right)$$

examples suffice. This is the **Occam bound**, and its proof takes a few lines (Section 9). The number of examples grows with $\ln |\mathcal{H}|$, the number of bits needed to name a hypothesis. For the class of all Boolean functions on n bits the bound is exponential in n : the no-free-lunch phenomenon reappears as a statement about sample size.

The Occam bound says nothing about computation time, and time can be the binding constraint. Under standard cryptographic assumptions, deterministic finite automata cannot be learned efficiently from random examples. Yet Dana Angluin’s L^* algorithm learns them exactly in polynomial time if the learner may ask whether particular strings belong to the language and may propose an automaton to be corrected with a counterexample (Section 9). What a learner can learn efficiently depends on the channel through which it gets information. The theme recurs throughout the course: pretraining on text supplies random examples, supervised fine-tuning supplies chosen demonstrations, preference learning supplies comparisons, and verifiable rewards supply something like answers to queries.

Simplicity, prediction, and compression

Ray Solomonoff proposed an idealized solution to induction in 1964: give every program for a universal machine a prior weight that halves with each additional bit of its length, and predict by conditioning the resulting prior on what has been seen. The predictor is uncomputable but optimal in a strong sense (Section 9), and it makes precise an idea that runs through this course: good prediction and short description are two faces of the same thing.

The practical version of that idea is the correspondence between probability and code length. Given any probability distribution q over messages, there is a prefix code, arithmetic coding, that encodes a message x in at most $-\log_2 q(x) + 2$ bits. Conversely, by Kraft’s inequality, the code lengths $\ell(x)$ of any prefix code define weights $2^{-\ell(x)}$ that sum to at most one, so every code is implicitly a probability model. If messages are drawn from a source p , the expected code length when coding with q is essentially the cross-entropy

$$H(p, q) = - \sum_x p(x) \log_2 q(x) = H(p) + D_{\text{KL}}(p \| q),$$

where $H(p)$ is the entropy of the source and D_{KL} is the Kullback-Leibler divergence, which is nonnegative and zero only when $q = p$. A model that predicts better compresses better, and the gap to the best possible compression is exactly the KL divergence. The minimum description length principle turns this into a method for choosing models: prefer the hypothesis H minimizing the length of H ’s own description plus the length of the data coded with H ’s help. Rissanen, 1978.

For independent examples, maximizing the likelihood of the data is the same as minimizing the total log loss. For a model of a distribution $q_{\theta}(x)$ fitted to samples from p_* , the expected log loss is the cross-entropy, so

$$\mathbb{E}_{x \sim p_*}[-\log q_{\theta}(x)] = H(p_*) + D_{\text{KL}}(p_* \| q_{\theta}).$$

The entropy $H(p_*)$ does not depend on θ , so in the limit of infinite data maximum likelihood minimizes the KL divergence from the true distribution to the model. This one identity connects the language-model training objective to compression and to the perplexity evaluations of Week 2.

Training a language model by minimizing next-token log loss is, in this sense, training a compressor for text. Whether compression of the written record amounts to understanding is a question for Section 7.

6. Language: rules, distributions, and prediction

What a sentence is

A written sentence is at once a sequence of marks, a grammatical construction, an expression about something, and a move in an interaction. The history of thinking about language has treated these aspects separately, and the distinctions remain useful for thinking about what a language model does and does not capture. This section follows the first two, structure and statistics; Section 7 takes up meaning and use.

Rules and productivity

Pāṇini's *Aṣṭādhyāyī*, a Sanskrit grammar usually dated to the middle of the first millennium BCE, describes the forms of a language through a compact system of interacting rules, together with technical conventions for applying them. Pāṇini, *Aṣṭādhyāyī*, translated by Śrīśa Chandra Vasu, 1891. Its central insight is productivity: a finite description can determine an unbounded number of expressions, including ones never observed. Noam Chomsky made the same point the foundation of modern linguistics, describing a grammar as a finite device that generates exactly the grammatical sentences of a language, and classifying such devices by their power in what became the Chomsky hierarchy of regular, context-free, context-sensitive, and recursively enumerable languages. Chomsky, *Three Models for the Description of Language*, 1956. Week 2 develops this hierarchy and the automata that recognize each level.

Learning a language from examples

Children learn their first language from a finite sample of it, mostly without being told which sentences are ungrammatical. How this is possible was a defining question of twentieth-century linguistics. E. Mark Gold gave it a formal version in 1967, with a negative answer: if sentences may arrive in any order, no learning procedure is guaranteed to settle on the correct grammar from positive examples alone, even when the language is known to be regular. Gold, *Language Identification in the Limit*, 1967. The result was taken up in the debate over the **poverty of the stimulus**, Chomsky's argument that the evidence available to a child is too thin to determine a grammar without strong innate constraints on the possible grammars. Chomsky, review of Skinner's *Verbal Behavior*, 1959.

Gold's order of presentation is adversarial, however. When sentences are drawn at random from a probability distribution, the picture changes: James Horning showed in 1969 that probabilistic context-free grammars can be learned, in a probabilistic sense, from random positive examples. Frequencies carry information that a bare list of sentences does not. A construction that should appear often but never does is evidence that it is ungrammatical. Learning from random samples of text, with frequency as evidence, is exactly the setting of a language model.

Prediction and statistics

Claude Shannon's 1948 paper founding information theory treated English as a stochastic source. It generated text from a series of approximations to English: letters chosen at random, then independently with their English frequencies, then letters chosen given the previous one or two, then words chosen given the previous word. The samples look more like English as the context grows. Shannon, *A Mathematical Theory of Communication*, 1948. In 1951 Shannon estimated the entropy of English by asking people to guess the next letter of a text, and found about one bit per letter when a long context is available, with bounds between 0.6 and 1.3 bits. Shannon, *Prediction and Entropy of Printed English*, 1951. The n-gram models of Week 2 are Shannon's approximations, and next-token prediction is his guessing game played by a machine.

Chomsky’s 1957 *Syntactic Structures* contains the most famous argument against statistical approaches to language. The sentences “Colorless green ideas sleep furiously” and “Furiously sleep ideas green colorless” had presumably never occurred in any text, so, Chomsky argued, any statistical model of grammaticalness would treat them as equally remote from English, although only the first is grammatical. Fernando Pereira answered forty years later with a calculation: a simple aggregate bigram model, which groups words into classes and was trained on newspaper text, assigns the grammatical sentence about 200,000 times the probability of its reversal. Pereira, *Formal Grammar and Information Theory: Together Again?*, 2000. A model that shares statistical strength across similar words can assign sensible probabilities to sentences it has never seen. This is the central idea of Weeks 2 and 3, and Peter Norvig’s essay on the “two cultures” of language science is a readable account of the debate. Norvig, *On Chomsky and the Two Cultures of Statistical Learning*.

Distribution

Zellig Harris argued that the structure of a language can be discovered from where its elements occur relative to one another, before anything is known about what they mean. Harris, *Distributional Structure*, 1954. The *distribution* of an element, whether a sound, a part of a word, or a word, is the set of environments in which it occurs: the elements around it and the positions it takes. Grammar, on this view, is a compact summary of distribution. Grouping elements that occur in similar environments into classes, such as nouns and verbs, replaces an endless list of attested combinations with a few general statements. Harris then turned to meaning. *Oculist* and *eye-doctor* occur in almost the same environments and are synonyms; *oculist* and *lawyer* share some environments and not others, “the amount of meaning difference corresponding roughly to the amount of difference in their environments.” He was careful not to claim more: the contexts of a word cannot all be listed, and they do not define its meaning fully, but differences in distribution give a rough measure of differences in meaning. The method also reaches beyond single words. If one sentence contains *A*, an active verb, and *C*, and a neighboring sentence contains *C*, a verb, and *A*, the second verb is likely to be the passive of the first. Harris developed such regularities between sentence forms into a theory of transformations, an idea that his student Noam Chomsky made central to generative grammar.

J. R. Firth’s version became a slogan: “You shall know a word by the company it keeps.” Firth, 1957. The *distributional hypothesis* that grew out of this work underlies word embeddings, which represent a word by statistics of the contexts it occurs in (Week 2), and a language model trained to predict the next token is, in a sense, a learned summary of the distribution of every context in its training data. Distributional evidence is exactly what such a model can see. It is informative but not the same as meaning: “increase” and “decrease” occur in very similar contexts and mean opposite things.

Neural language models turned distributional statistics into learned representations. The neural probabilistic language model of Yoshua Bengio and colleagues learned a vector for each word jointly with a predictor of the next word, so that words used in similar contexts receive similar vectors and share statistical strength. Bengio, Ducharme, Vincent, and Jauvin, 2003. Later models compute a different vector for each occurrence of a word, depending on its context. Formal structure, frequency, and prediction illuminate different aspects of language, and Week 2 begins building models of each.

7. Meaning: reference, use, intent, and context

Reference and composition

Aristotle’s *On Interpretation* separates a name, which signifies by convention, from a statement, which can be true or false only once expressions are combined. Aristotle, *On Interpretation*. Gottlob Frege distinguished an expression’s reference (what it picks out) from its sense (the way it presents what it picks out): “the author of submission 17” and “the winner of the student prize” may refer to the same person while conveying different information. Frege, *On Sense and Reference*, 1892. The meaning of a whole depends on the meanings of its parts and how they are combined. “Every student read a paper” has two readings, $\forall x \exists y R(x, y)$ and $\exists y \forall x R(x, y)$, which differ in truth conditions while using the same words. A representation that records only which words occur cannot distinguish them.

Use and implicature

Ludwig Wittgenstein’s *Philosophical Investigations* examines language as a family of activities, such as giving orders, reporting, and asking, and suggests that for a large class of cases the meaning of a word is its use in the language. Wittgenstein, *Philosophical Investigations*, 1953. Paul Grice explained how speakers communicate more than they literally say. If asked whether all four experiments succeeded, the answer “Some succeeded” suggests that not all did, although it is logically consistent with all four succeeding; the suggestion can be cancelled by adding “in fact, all four did.” Grice, *Logic and Conversation*, 1975. An assistant that answers questions must handle this layer of meaning, which is invisible to a purely logical analysis of sentences.

Intent, context, and communication

Grice’s example suggests a picture of communication that is worth making explicit. A speaker has something to convey, an intent, and chooses an utterance that, together with the context the two parties share, should lead the listener to it. The listener reconstructs an intent from the utterance and from their own view of the context:

$$\text{intent} \xrightarrow[\text{speaker's context}]{\text{expression}} \text{utterance} \xrightarrow[\text{listener's context}]{\text{interpretation}} \text{inferred intent.}$$

The inferred intent need not equal the original, since the two sides may not share the relevant context. “It’s cold in here” can report the temperature, explain why someone is wearing a coat, or ask someone to close the window. Which it does depends on the physical situation, on what was said before, and on what the listener believes about the speaker. The words fix much less than the meaning.

One useful description of this picture is that communication is a context-dependent, lossy compression of intended meaning: speakers leave most of what they mean unsaid and rely on listeners to supply it from shared knowledge. This is a model, not a definition of language or a claim about why language evolved, but it has been made precise in particular settings. Michael Frank and Noah Goodman model a listener who infers the intended referent by reasoning about which utterance a cooperative speaker would have chosen, and the model predicts people’s interpretations in simple reference games. Frank and Goodman, 2012. Noga Zaslavsky, Charles Kemp, Terry Regier, and Naftali Tishby show that the color-naming systems of many languages lie close to the optimal tradeoff between the complexity of a vocabulary and the accuracy with which it conveys a color. Zaslavsky, Kemp, Regier, and Tishby, 2018.

Two senses of compression should be kept apart. **Text compression**, from Section 5, is a theorem about probability and code length: a better model of text gives shorter codes. **Communicative compression** is an explanatory model of how a short utterance can convey a rich intent, given context. A language model is trained on text alone. Its objective rewards predicting what a writer wrote next, not recovering what the writer meant, and no label for the intent is ever supplied. The text nevertheless carries evidence about intent and context, often a great deal of it. How much is recoverable, and what requires grounding in the world or interaction with people, are the questions of the rest of this section.

Form, meaning, and statistics

Shannon’s theory of communication, the source of the statistical view of Section 6, deliberately set meaning aside: messages frequently have meaning, but “these semantic aspects of communication are irrelevant to the engineering problem.” Shannon, *A Mathematical Theory of Communication*, 1948. Entropy measures uncertainty about which message is sent, not what it says. A language model inherits this stance: its training objective is predictive, and it is rewarded for assigning high probability to observed text whether or not that text is true.

Emily Bender and Alexander Koller turned this into an argument that meaning cannot be learned from form alone. In their thought experiment, two people stranded on separate islands communicate through an underwater cable. A hyperintelligent octopus taps the cable, learns the statistical patterns of their messages, and eventually impersonates one of them convincingly. Then one of the islanders is chased by an angry bear, grabs a couple of sticks, and asks the other for advice on defending herself. The octopus cannot help, because nothing in the patterns of the messages connects words to the objects and situations they are about. Bender and Koller, *Climbing towards NLU*, 2020. This is a version of the older symbol grounding problem: how can the symbols of a formal system mean anything to the system itself, rather than only to the people who interpret it? Harnad, *The Symbol Grounding Problem*, 1990.

The opposing view holds that much of meaning is relational. Steven Piantadosi and Felix Hill argue that the meaning of a concept consists in large part of its relations to other concepts, its conceptual role, and that a model trained on text can learn such relations even without reference to the world. Piantadosi and Hill, *Meaning without Reference in Large Language Models*, 2022. The question is not settled, and this course does not settle it. It does suggest a more precise formulation. A distribution over text is a very rich object: its conditional structure encodes grammar, facts stated in text, and the patterns of reasoning people write down. The question is what information about the world, and about communicative intent, is recoverable from that distribution, and what additional information comes from the later stages of training, in which humans demonstrate, compare, and check.

A final result shows that even perfect statistical modeling has consequences one might not want. Adam Tauman Kalai and Santosh Vempala consider arbitrary facts, such as which paper a person wrote in a given year, whose truth cannot be inferred from patterns in other facts. They prove that a language model that is calibrated, in the sense that its probabilities match frequencies, must generate false statements of this kind at a rate roughly equal to the fraction of such facts that appear exactly once in its training data, a Good-Turing estimate of the unseen mass. Kalai and Vempala, *Calibrated Language Models Must Hallucinate*, 2024. The result has a precise scope. It is a lower bound for calibrated models on facts of this arbitrary kind, and it gives no statistical reason to hallucinate facts that appear many times in the data or that follow systematic patterns, such as arithmetic. For the facts it does cover, hallucination is a consequence of calibrated prediction rather than an engineering defect, so reducing it means giving up some calibration, which post-training (Week 7) can do, or consulting an external source such as retrieved documents.

8. Hands-on: an automatic-differentiation engine

Every neural model in this course is trained by gradient methods, and every gradient is computed by reverse-mode automatic differentiation. The lecture’s hands-on segment works through an engine that does this, written from scratch in about 300 lines of NumPy: `tinyml/autograd.py`. Part A of the Week 1 notebook follows this section step by step; Part B measures the cost of gradients and trains two classifiers with the engine.

Tensors and computation graphs

A **tensor** here is an array together with what is needed to differentiate through it: its data, a gradient of the same shape that the backward pass fills in, the tensors it was computed from, and a rule for passing gradients back to them. Evaluating an expression such as $L = \text{mean}((XW + b)^2)$ builds a directed acyclic graph whose leaves are the inputs and parameters and whose root is the scalar loss.

The backward pass

Backpropagation computes the gradient of a scalar loss with respect to every parameter of a composed computation by applying the chain rule backwards through the graph. Each operation contributes one local rule. If an operation maps an input u to an output z , and $\bar{z} = \partial L / \partial z$ is already known, the rule returns $\bar{u} = (\partial z / \partial u)^\top \bar{z}$, a **vector-Jacobian product**. The backward pass visits the graph in reverse topological order, so that each node is processed only after every node that uses it, and applies these rules from the root to the leaves.

Two details carry most of the subtlety. First, when a value is used more than once, its gradient is the **sum** of the contributions from its uses; this is the multivariable chain rule. In $f(x, y) = \log(1 + e^{xy}) + x^2$ the variable x is used twice, and

$$\frac{\partial f}{\partial x} = y \sigma(xy) + 2x, \quad \sigma(t) = \frac{1}{1 + e^{-t}},$$

one term for each use. Second, **broadcasting is reuse**. In $Z = XW + b$, with X of shape $n \times d$ and W of shape $d \times k$, the bias $b \in \mathbb{R}^k$ is added to each of the n rows, so its gradient is the sum of the rows of $G = \partial L / \partial Z$. The matrix product has the rules $\partial L / \partial W = X^\top G$ and $\partial L / \partial X = GW^\top$. In every case the gradient has the shape of the tensor it belongs to, which is a useful check when deriving rules by hand.

Checking gradients

A mistake in a backward rule rarely causes a crash. It silently produces a wrong gradient, and training quietly does worse. Every rule in the engine is therefore checked against the central difference

$$\frac{\partial f}{\partial x_i} \approx \frac{f(x + he_i) - f(x - he_i)}{2h}.$$

Its truncation error is of order h^2 , while rounding in the two evaluations of f contributes about ϵ_{mach}/h , so the total is smallest near $h \approx \epsilon_{\text{mach}}^{1/3} \approx 10^{-5}$ in double precision. The engine's tests also compare its gradients with those of PyTorch.

Why reverse mode

Reverse mode has a remarkable complexity property: the gradient of a function with respect to all of its inputs can be computed with only a small constant factor more arithmetic than evaluating the function once. Baur and Strassen, 1983. Finite differences, by contrast, need two evaluations of the function for every input, which for a network with tens of thousands of parameters means minutes instead of milliseconds, as Part B of the notebook measures. This cheap gradient principle is why networks with billions of parameters can be trained at all.

Gradients and updates

Backpropagation computes derivatives; the **optimizer** decides how to use them. Gradient descent (Section 5) moves each parameter a small step against its gradient; the variants of Week 4 adapt the step using the history of gradients. Because gradients accumulate across backward passes, a training loop clears them before each step, then runs the forward pass, the backward pass, and the update. The same loop, with a cross-entropy loss and a small network, trains the classifiers in Part B of the notebook.

From here to language models

A language model applies these ingredients to the distribution over strings of Section 2. Each conditional $q_\theta(x_t \mid x_{<t})$ is a classifier over the vocabulary, computed by a network from the prefix; training minimizes the log loss of observed next tokens, which is self-supervised; and generation samples one token at a time, feeding each back as input. The lifecycle of a modern model has four stages, each with its own source of learning signal:

Stage	Signal	What changes	Weeks
Pretraining	Next tokens of a large text corpus	All parameters, from a random start	2-4
Supervised finetuning	Demonstrations of desired behavior	Parameters, from the pre-trained model	5, 7
Preference and RL post-training	Comparisons and rewards	Parameters, relative to a reference model	6, 7
Inference	None: prompts, retrieved text, and tool outputs condition a fixed model	Nothing in the parameters	5, 8

The last row is a distinction worth keeping in mind from the start. A model can appear to learn during a conversation, because a fact placed in its context changes its predictions. That is conditioning, not learning in the sense of Section 3: nothing persists once the context is gone.

Next week

We now have the vocabulary to ask precise questions about the model we will build. Week 2 starts with the simplest language models and three questions: what makes a string belong to a language, how to assign probabilities to strings, and how to measure whether a predictor is any good. We answer them first with grammars and counts.

9. Deeper reading

This section collects the formal results that Sections 4 and 5 describe in words. The lecture does not cover them.

Universal intelligence

Shane Legg and Marcus Hutter define the universal intelligence of an agent π as its expected performance across all computable environments, weighted by simplicity:

$$\Upsilon(\pi) = \sum_{\mu \in E} 2^{-K(\mu)} V_{\mu}^{\pi}.$$

Here E is the class of computable reward-generating environments, V_{μ}^{π} is the expected total reward the agent obtains in environment μ , and $K(\mu)$ is the Kolmogorov complexity of μ : the length of the shortest program that implements it on a fixed universal machine. Legg and Hutter, 2007. The definition is elegant and explains a common intuition, that intelligence is the ability to do well across many environments with a bias toward simple ones. It is also uncomputable, since K is uncomputable and the sum ranges over all computable environments, so it cannot be used as a benchmark directly.

The Occam bound

Here is the proof of the Occam bound of Section 5. Suppose the learner searches a finite hypothesis class $\mathcal{H} \supseteq \mathcal{C}$ and outputs any hypothesis that agrees with every training example. Call a hypothesis bad if its error exceeds ε . A fixed bad hypothesis agrees with one random example with probability at most $1 - \varepsilon$, so it agrees with all m independent examples with probability at most

$$(1 - \varepsilon)^m \leq e^{-\varepsilon m}.$$

There are at most $|\mathcal{H}|$ bad hypotheses, so by the union bound the probability that any of them survives is at most $|\mathcal{H}|e^{-\varepsilon m}$. Requiring this to be at most δ gives the sample size

$$m \geq \frac{1}{\varepsilon} \left(\ln |\mathcal{H}| + \ln \frac{1}{\delta} \right).$$

This is the **Occam bound**: the number of examples needed grows with the logarithm of the number of hypotheses, that is, with the number of bits needed to name a hypothesis. For example, with $|\mathcal{H}| = 2^{20}$, $\varepsilon = 0.05$, and $\delta = 0.01$, the bound requires $m \geq 20(20 \ln 2 + \ln 100)$, about 370 examples. If $\mathcal{H}$ contains all Boolean functions on n input bits, then $\ln |\mathcal{H}| = 2^n \ln 2$, and the bound becomes exponential in n : the no-free-lunch phenomenon reappears as a sample-complexity statement. Infinite classes are handled by replacing $\ln |\mathcal{H}|$ with a combinatorial measure of complexity such as the VC dimension. The general theory, including the case where no hypothesis fits the data perfectly, is developed in Shalev-Shwartz and Ben-David, *Understanding Machine Learning* and in Chapter 3 of the course's AI and Machine Learning companion.

Hardness and queries

The Occam bound says nothing about time. Finding a hypothesis consistent with the data can be computationally hard even when few examples suffice, and for some classes no efficient learner exists at all under standard cryptographic assumptions. Michael Kearns and Valiant showed that deterministic finite automata cannot be efficiently PAC learned, even if the learner may output any polynomially evaluable hypothesis, unless certain cryptographic problems (such as inverting RSA) are easy. Kearns and Valiant, 1994. The intuition is that a finite automaton can compute a pseudorandom-looking function; if one could learn it from random examples, one could predict the output of a cryptographic generator.

The hardness depends on how the learner obtains information. Dana Angluin's L^* algorithm learns any regular language exactly, in time polynomial in the number of states of its minimal automaton and the length of the longest counterexample it receives, if the learner may ask two kinds of question: membership queries ("is this string in the language?") and equivalence queries ("is this automaton correct? if not, give me a counterexample"). Angluin, 1987. The contrast between the two results is the one drawn in Section 5: what a learner can learn efficiently depends on the channel through which it gets information.

These results concern automata, not neural networks trained by gradient descent, and they are worst-case statements. They should be read as warnings about what cannot be guaranteed in general, not as predictions about what happens on natural language. The gap between worst-case hardness and empirical success recurs throughout the course.

Solomonoff induction

Ray Solomonoff proposed an idealized solution to induction in 1964. Consider all programs for a fixed universal machine, and give each program p prior weight $2^{-|p|}$, so that shorter programs are more probable. The universal prior probability of a finite string x is the total weight of programs whose output begins with x . Predict the next symbol by conditioning this prior on what has been seen. Solomonoff, *A Formal Theory of Inductive Inference*, Part I, 1964. Solomonoff later proved that this predictor is optimal in a strong sense: for any computable data source μ , its total expected prediction error over an infinite sequence is bounded by a constant proportional to the complexity $K(\mu)$ of the source. Solomonoff induction is uncomputable, like the Legg-Hutter measure built on it, but it makes precise an idea that runs through this course: good prediction and short description are two faces of the same thing.

10. Background reading

The basics of Sections 3, 5, and 8 are developed in full elsewhere. The table lists where. “Companion” refers to the course’s *AI and Machine Learning* companion; the module chapters refer to the Background Notes on Machine Learning and AI.

Topic	Companion	Background Notes	Textbook
Terminology; the objects being learned	Chapter 1	Foundations 1	Russell and Norvig, Chapters 1-2
Linear algebra and calculus	Chapter 2	Foundations 2, 3	Goodfellow et al., Chapters 2, 4
Probability, likelihood, and Bayes’ rule	Chapter 5	Foundations 4; ML 4, 5	Goodfellow et al., Chapter 3
Risk, generalization, and PAC learning	Chapter 3	Foundations 5; ML 7	Shalev-Shwartz and Ben-David, Chapters 2-6
Information theory	Chapter 6	Foundations 5	Cover and Thomas, Chapter 2
Evaluation and data leakage	Chapter 7	ML 6	
Neural networks and back-propagation	Chapter 10	DL 1; Foundations 3	Goodfellow et al., Chapter 6
Optimization and regularization	Chapter 11	DL 3, 5	Goodfellow et al., Chapters 7-8
Search, logic, and probabilistic reasoning		AI 1-9	Russell and Norvig, Parts II-IV
Numerical computing with arrays		Foundations 6	

For reading in the order of this lecture: Turing’s 1950 paper is short and still rewarding; Valiant’s 1984 paper introduces the PAC model in eight pages; Chapters 2 and 3 of Shalev-Shwartz and Ben-David develop the learning theory; and the companion’s Chapters 1, 3, 5, and 7 cover the basics at the level assumed in Weeks 2 through 4.

11. Exercises

Exercise 1: Sources of learning signal

For each situation, identify the task, the source of the learning signal, and what changes in the model.

1. A Transformer predicts masked words from the surrounding text.
2. A pretrained model is finetuned on instruction-answer pairs written by people.
3. A fixed model receives retrieved documents before answering a question.
4. A model samples programs, receives unit-test results, and updates its parameters to make passing programs more likely.

Exercise 2: Bayes' rule

A model assigns prior probability 0.2 to spam. The word “prize” appears in spam with probability 0.6 and in other email with probability 0.05.

1. Compute the probability of spam given that an email contains “prize”.
2. Which quantities does a generative classifier estimate here, and which does a discriminative classifier estimate instead?
3. Give one example each of statistical inference, probabilistic inference, and deployment inference in this setting.

Exercise 3: Definitions of intelligence

1. Why is the Legg-Hutter measure incomputable? Name the two separate sources of incomputability.
2. Give an example of a system that would score well on a fixed benchmark of tasks but poorly under Chollet’s skill-acquisition definition, and explain the difference.
3. Turing’s prediction for the year 2000 was stated in terms of storage (10^9 bits) rather than computation. What quantities would you use today to describe the resources of a language model, and why?

Exercise 4: The Occam bound

1. A consistent learner searches a class of 2^{20} hypotheses. How many examples guarantee error at most 0.05 with probability at least 0.99? How does the answer change if the class has 2^{40} hypotheses, and if the required error is halved?
2. Let $\mathcal{H}$ be all Boolean functions on n bits. Evaluate the Occam bound and explain its relation to the no-free-lunch example of Section 5.
3. The derivation used $(1 - \varepsilon)^m \leq e^{-\varepsilon m}$ and a union bound. Which step uses the assumption that training and test inputs come from the same distribution?

Exercise 5: Logistic regression and generalization

Use $q_w(y = 1 \mid x) = \sigma(w^\top x)$ with log loss.

1. Starting from $w = (0, 0)$, take one gradient step with $\eta = 0.1$ on the example $x = (2, 1)$, $y = 1$. Report the loss before and after.
2. Evaluate the loss on a second example $x' = (-2, 1)$, $y' = 1$ before and after the same step. Explain why an update that helps one example does not demonstrate generalization.
3. Classify each of the following as a parameter, a hyperparameter, or an evaluation choice: w ; the learning rate; the number of steps; holding out every email from particular senders.

Exercise 6: Prediction and compression

Let a source emit symbols $\{A, B, C\}$ with probabilities $p = (1/2, 1/4, 1/4)$.

1. Compute the entropy $H(p)$ in bits and give a prefix code that achieves it.
2. A model uses the uniform distribution $q = (1/3, 1/3, 1/3)$. Compute the cross-entropy $H(p, q)$ and the KL divergence $D_{\text{KL}}(p\|q)$ in bits, and interpret the difference as extra code length.
3. Explain why minimizing the average log loss of a language model on held-out text is the same as minimizing the length of an ideal code for that text built from the model.

Exercise 7: Meaning and its evidence

1. Compare “Mira admires Arun” with “Arun admires Mira.” What does a bag-of-words representation lose? Give a relation interpreting “admires” under which one sentence is true and the other false.
2. Explain why “Some succeeded” suggests that not all succeeded without entailing it. Give a continuation that cancels the suggestion.
3. “Increase” and “decrease” have similar distributions. Describe a prediction task on which a model that confuses them would fail.
4. Summarize, in a paragraph, the strongest version of the argument that a model trained only on text cannot learn meaning, and the strongest reply. What experiment, if any, could distinguish the two positions?
5. Harris’s test on a toy corpus. Take the six sentences “the oculist examined the eye”, “the eye-doctor examined the eye”, “the oculist prescribed glasses”, “the eye-doctor prescribed glasses”, “the lawyer examined the contract”, and “the lawyer drafted the contract”. Represent each of *oculist*, *eye-doctor*, and *lawyer* by the counts of the word that follows it, and compute the cosine similarity of each pair. Then add the sentence “the oculist is an eye-doctor” and recompute. Why does a sentence that asserts the synonymy make the two words look less alike, and how does this match Harris’s remark that synonyms share environments “except chiefly for sentences which contain both”?
6. Describe three situations in which “It’s cold in here” is used, in turn, to report a fact, to explain an action, and to make a request. For each, say what information the listener needs in order to recover the intent, and whether that information would appear in a transcript of the conversation.

Exercise 8: Gradients by hand

1. For $f(x, y) = \log(1 + e^{xy}) + x^2$, draw the computation graph and compute $\partial f/\partial x$ and $\partial f/\partial y$ at $(x, y) = (1.5, -0.5)$ by a backward pass through it. Identify the two contributions to $\partial f/\partial x$, and check your answers against Section 2 of the Week 1 notebook.
2. Let X be $n \times d$, W be $d \times k$, and $b \in \mathbb{R}^k$, and let $Z = XW + \mathbf{1}b^\top$ and $L = \frac{1}{nk} \sum_{i,j} Z_{ij}^2$. Derive $\partial L/\partial W$ and $\partial L/\partial b$, and check that each has the shape of the variable it belongs to.
3. For $v \in \mathbb{R}^d$ and $L = \|Xv\|^2$, show that $\nabla_v L = 2X^\top Xv$. (This is the exercise at the end of Part A of the notebook.)
4. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is computed by a program with T arithmetic operations. How many evaluations of f does it take to estimate the gradient by central differences, and how much arithmetic in total? Compare with one backward pass.

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